



Hamed Hosseini

AI ENGINEERING LEAD · INTELLIGENT SYSTEMS

Tehran, Iran · hosseini.hamed@ut.ac.ir

LinkedIn: [linkedin.com/in/hamed-hosseini](https://www.linkedin.com/in/hamed-hosseini) · GitHub: github.com/Hamed-hosseini

AI and data science leader combining hands-on engineering with research depth in robotics and computer vision. Experienced in leading multidisciplinary teams and building applied machine learning, fraud analytics, language and document intelligence, and scalable data products.

LEADERSHIP PROFILE

AI product architecture · Data science team leadership · Applied machine learning · NLP and OCR · LLM applications · Data platforms and streaming · Robotics and computer vision

PROFESSIONAL EXPERIENCE

Data Science Team Lead · Banking Solutions

Pouya Co. · February 2018–present

- Built and led a five-person AI product team; mentored three junior data scientists.
- Co-designed modular, microservice-based AI products with an emphasis on maintainability and scale.
- Led online and offline card-fraud detection using behavior modeling, statistical inference, generative models, and anomaly detection.
- Oversaw an LLM-assisted customer service system integrating semantic analysis, NLP, OCR, LangChain, and LangGraph.
- Designed scalable data pipelines using Spark, Kafka, and Hadoop to support monitoring and analytics.
- Developed customer credit evaluation and segmentation solutions; contributed to image-analysis and face-detection systems.

Lecturer · Artificial Intelligence in Robotics and Mechatronics

University of Tehran · August 2023–present

Teaching AI for robotics and mechatronics, alongside academic and lab research.

Research Assistant · Human–Robot Interaction Laboratory

University of Tehran · August 2017–present

Robotic grasping, deep learning, geometric scene understanding, simulation, and real-robot validation.

Research Assistant · Parsian Robotic Lab

Amirkabir University of Technology · December 2015–August 2017

Small-size robot soccer strategy, control, and team coordination; competition experience at IranOpen and RoboCup.

SELECTED APPLIED AI WORK

Agentic assistant and enterprise workflows

Designed an AI-driven assistant using open-source and API-based LLMs for coding, reporting, and project management.

Financial intelligence

Credit scoring and segmentation, online/offline fraud detection, NLP/OCR-enabled customer support, and data analytics.

Robotics and computer vision

Deep-learning-based grasp detection, object/scene understanding, image captioning and translation, and robotic simulation.

EDUCATION

Ph.D. · Artificial Intelligence and Robotics

2017–2025 · University of Tehran · GPA 17.25/20

Thesis: Intelligent Robotic Grasp Using Deep Neural Networks.

M.Sc. · Electrical Engineering, Control

2015–2017 · Amirkabir University of Technology · GPA 17.30/20

Thesis: Employing Biological Signals in the Rehabilitation Procedure.

B.Sc. · Electrical Engineering

2011–2015 · Amirkabir University of Technology · GPA 16.12/20

Thesis: Artifact Detection at the EEG Signals.

TECHNICAL SKILLS

Languages	Python, C++, MATLAB, R
AI/ML	PyTorch, TensorFlow, Keras, scikit-learn, deep learning, anomaly detection
LLM systems	LLM applications, LangChain, LangGraph, NLP, OCR
Data platforms	Apache Spark, Kafka, Hadoop, SQL, Oracle, Redis, MongoDB
Robotics	ROS, Gazebo, CoppeliaSim, PyBullet, computer vision
Engineering	Microservices, Git, model development, analytics pipelines

SELECTED RESEARCH PUBLICATIONS

- Multi-Modal Robust Geometry Primitive Shape Scene Abstraction for Grasp Detection, *IEEE Access* (2024).
- AGILE: Approach-Based Grasp Inference Learned from Element Decomposition, *ICRoM* (2023).
- Embodied AI for Kitchen Scene Rearrangement, *Intelligent Service Robotics* (2026).

RECOGNITION

Ranked 14th in Alibaba's ZhejiangLab Cup Global AI Competition, Zero-shot Learning Picture Recognition (2018; 3,225 teams). RoboCup Small Size League: 4th place (2017). IranOpen Small Size League: 1st technical challenge and 3rd overall (2017).

Academic publication profile: [Google Scholar](#) · Contact: hosseini.hamed@ut.ac.ir